SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks: 50**

84%

Annexure A

Scenario-Based

Code of Conduct:

I S.K. Imran (192525348) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses.**
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation.**
2. Expand the compressed address back into its **full hexadecimal representation.**
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry.**
3. The **next-hop network/interface** used to forward the packet.

4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range.**
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context(2)	Shows good understanding with minor gaps(1.5)	Partial understanding; misses key aspects(1)	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved(2)	Identifies most key issues with minor omissions(1.5)	Identifies limited issues; some irrelevant points(1)	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario(2.5)	Applies concepts with minor errors(2)	Limited or partially correct application(1.5)	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale(2)	Makes reasonable decisions with some justification (1.5)	Makes weak decisions with limited justification(1)	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented(1.5)	Generally clear with minor issues(1)	Somewhat unclear or poorly structured(0.5)	Disorganized and difficult to understand

	understanding of Scenario (20%)	Identification of key issue (20%	Application of Concepts (25%)	Decision making (20%)	Clarity of Response (15%)	
Q1	2	2	2.5	2	1.5	10
Q2	2	2	2.5	2	1.5	10
Q3	1.5	1.5	2	1	1	7
Q4	2	2	2	1	1	8
Q5	1.5	2	1	1	1.5	7
						42

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Scenario-Based

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Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

Given IPv6 Address:
`2001:0db8:0000:0000:0000:ff00:0042:8329`

1. Compressed IPv6 Address

IPv6 compression rules:

- Remove leading zeros in each group.
- Replace consecutive groups of `0000` with `::` only once.

Compressed Address:

`2001:db8::ff00:42:8329`

2. Expanded IPv6 Address

The compressed form contains `::`, which represents the missing zero groups.

Expanded Address:

```
2001:0db8:0000:0000:0000:ff00:0042:8329
```

3. Network Prefix for /64

An IPv6 address has **128 bits**.

With a **/64 prefix**, the first **64 bits** represent the network portion.

Since each IPv6 group contains **16 bits**:

```
64 ÷ 16 = 4 groups
```

Therefore, the network portion is:

```
2001:0db8:0000:0000
```

Network Prefix:

```
2001:db8::/64
```

4. Number of Possible Host Addresses

Total IPv6 bits = **128**

Network bits = **64**

Host bits:

```
128 − 64 = 64 bits
```

Total possible host addresses:

2^{64} = 18,446,744,073,709,551,616

Final Answer Table

Item	Answer
Given IPv6 Address	`2001:0db8:0000:0000:0000:ff00:0042:8329`
Compressed Form	`2001:db8::ff00:42:8329`
Expanded Form	`2001:0db8:0000:0000:0000:ff00:0042:8329`
Prefix Length	`/64`
Network Prefix	`2001:db8::/64`

Item	Answer
Host Bits	64 bits
Total Possible Host Addresses	2^{64}
Number of Host Addresses	18,446,744,073,709,551,616

Conclusion

The IPv6 address is compressed as **`2001:db8::ff00:42:8329`**. Using a **/64 prefix**, the network prefix is **`2001:db8::/64`**. The remaining **64 bits** are available for host addressing, giving a total of **2^{64} = 18,446,744,073,709,551,616 possible addresses**.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

Here's the full subnetting breakdown for **10.10.0.0/16**, allocating blocks sequentially in the order given (Software Dev → QA → HR → Executive Management):

1–3. Network, Broadcast, Host Range & Usable Hosts

Department	Subnet	Network Address	Broadcast Address	Usable Host Range	Usable Hosts
Software Development	10.10.0.0/24	10.10.0.0	10.10.0.255	10.10.0.1 – 10.10.0.254	254
Quality Assurance	10.10.1.0/25	10.10.1.0	10.10.1.127	10.10.1.1 – 10.10.1.126	126
Human Resources	10.10.1.128/26	10.10.1.128	10.10.1.191	10.10.1.129 – 10.10.1.190	62
Executive Management	10.10.1.192/27	10.10.1.192	10.10.1.223	10.10.1.193 – 10.10.1.222	30

How each was derived

- **/24** → subnet mask 255.255.255.0 → block size 256 → hosts = $2^8 - 2 = 254$
- **/25** → subnet mask 255.255.255.128 → block size 128 → hosts = $2^7 - 2 = 126$
- **/26** → subnet mask 255.255.255.192 → block size 64 → hosts = $2^6 - 2 = 62$
- **/27** → subnet mask 255.255.255.224 → block size 32 → hosts = $2^5 - 2 = 30$

Each new subnet starts right after the previous one's broadcast address, so there's no overlap and no wasted address in between the four allocated blocks themselves.

4. Unused (Remaining) Address Space

Total allocated so far: 10.10.0.0 – 10.10.1.223 (out of the full /16 range 10.10.0.0 – 10.10.255.255).

Unused range: 10.10.1.224 – 10.10.255.255

Breaking that down:

- **10.10.1.224 – 10.10.1.255** → a leftover 32-address chunk right after Executive Management (could become another /27 subnet)
- **10.10.2.0 – 10.10.255.255** → 254 full /24-sized blocks (over 65,000 addresses) still completely free for future departments or expansion

Quick summary

- Total addresses in 10.10.0.0/16 = 65,536
- Allocated to the four departments = 256 + 128 + 64 + 32 = 480 addresses
- Remaining unused = 65,536 − 480 = **65,056 addresses** available for future growth

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

Given IPv6 Address

2001:0db8:abcd:0000:0000:0000:0000:0000/64

1. Compressed IPv6 Notation

IPv6 compression rules:

- Remove leading zeros in each block.
- Replace consecutive groups containing only `0000` with `::` once.

Therefore:

2001:db8:abcd::/64

2. Full Hexadecimal Representation

Expanding the compressed address back to 8 groups of 4 hexadecimal digits:

2001:0db8:abcd:0000:0000:0000:0000:0000

3. Network Prefix Using `/64`

An IPv6 address contains **128 bits**.

With a **/64 prefix**, the first **64 bits** represent the network portion.

Network Prefix:

2001:0db8:abcd:0000::/64

The remaining **64 bits** are used for host/interface addresses.

4. Total Number of Host Addresses

Host bits:

128 − 64 = 64 bits

Therefore:

Total host addresses = 2^{64}

= 18,446,744,073,709,551,616 addresses

So, the subnet can theoretically support:

18,446,744,073,709,551,616 host/interface addresses.

5. Total Possible Host Addresses Available Within the Network

Since the prefix length is `/64`, there are **64 bits available for host addressing**.

264=18,446,744,073,709,551,616

Final Answer: The IPv6 `/64` network provides **18,446,744,073,709,551,616 possible addresses**.

Summary

Item	Answer
Given Address	`2001:0db8:abcd:0000:0000:0000:0000:0000/64`
Compressed Address	`2001:db8:abcd::/64`
Full Address	`2001:0db8:abcd:0000:0000:0000:0000:0000`
Network Prefix	`2001:0db8:abcd:0000::/64`
Host Bits	`64 bits`
Total Possible Host Addresses	2^{64}
Total Number of Addresses	**18,446,744,073,709,551,616**

Conclusion: The smart hospital can use a `/64` IPv6 subnet to support an extremely large number of medical devices, IoT sensors, computers, and other connected devices.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

Given Routing Table

Network Address	Prefix Length
`192.168.1.0`	`/24`
`192.168.2.0`	`/24`
`192.168.3.0`	`/24`

Destination IP Address

192.168.2.45

1. Destination Subnet

A `/24` subnet uses the subnet mask:

255.255.255.0

For the destination IP:

`192.168.2.45`

The first three octets represent the network portion. Therefore, the packet belongs to:

192.168.2.0/24

2. Matching Routing Table Entry

The router compares the destination IP address with its routing table.

`192.168.2.45` matches:

192.168.2.0/24

Therefore, the matching routing table entry is:

192.168.2.0/24

3. Next-Hop Network/Interface

The packet will be forwarded through the **router interface or next-hop associated with the `192.168.2.0/24` route**.

Therefore:

Next-hop network/interface → 192.168.2.0/24 interface

Note: The exact interface name or next-hop IP address cannot be determined because it is not provided in the routing table.

4. Default Route

If the destination IP address does **not** match any routing table entry, the router uses the **default route**:

0.0.0.0/0

This route forwards packets to the ISP's configured **default gateway or next-hop router**.

Question	Answer
Destination IP	`192.168.2.45`
Destination Subnet	**192.168.2.0/24**
Matching Route	**192.168.2.0/24**
Next-Hop/Interface	**Interface connected to `192.168.2.0/24`**
Default Route	**0.0.0.0/0**

Conclusion: Since `192.168.2.45` belongs to the `192.168.2.0/24` network, the router matches the `192.168.2.0/24` routing entry and forwards the packet through the corresponding interface or next hop. If no matching route exists, the router uses the default route `0.0.0.0/0`.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

Given Information

- **Administration LAN:** `192.168.10.0/26`
- **Finance LAN:** `192.168.10.64/26`
- **Administration Gateway:** `192.168.10.1`
- **Finance Gateway:** `192.168.10.65`

A `/26` subnet has the subnet mask:

255.255.255.192

Each `/26` subnet contains:

2(32−26)=26=64 total addresses

Out of these, **62 addresses are usable for hosts**.

1. Network Address and Broadcast Address

Administration LAN – `192.168.10.0/26`

- **Network Address:** `192.168.10.0`
- **Broadcast Address:** `192.168.10.63`

Finance LAN – `192.168.10.64/26`

- **Network Address:** `192.168.10.64`
- **Broadcast Address:** `192.168.10.127`

2. Valid Host IP Address Range

Administration LAN

Network: `192.168.10.0/26`

- First usable host: `192.168.10.1`
- Last usable host: `192.168.10.62`

Valid Host Range: `192.168.10.1 – 192.168.10.62`

`192.168.10.1` is used as the default gateway.

Finance LAN

Network: `192.168.10.64/26`

- First usable host: `192.168.10.65`
- Last usable host: `192.168.10.126`

Valid Host Range: `192.168.10.65 – 192.168.10.126`

`192.168.10.65` is used as the default gateway.

3. Suitable IP Addresses for Three Computers

Administration LAN

Computer	Suggested IP Address
Computer 1	`192.168.10.2`
Computer 2	`192.168.10.3`
Computer 3	`192.168.10.4`

Subnet Mask: **255.255.255.192**

Finance LAN

Computer	Suggested IP Address
Computer 1	`192.168.10.66`
Computer 2	`192.168.10.67`
Computer 3	`192.168.10.68`

Subnet Mask: **255.255.255.192**

4. Default Gateway for Each LAN

LAN	Default Gateway
Administration LAN	**192.168.10.1**
Finance LAN	**192.168.10.65**

The default gateway is the IP address of the router interface connected to that LAN. Hosts use it to communicate with devices located on other networks.

Final Answer Summary

LAN	Network Address	Valid Host Range	Broadcast Address	Default Gateway
Administration	`192.168.10.0/26`	`192.168.10.1 – 192.168.10.62`	`192.168.10.63`	`192.168.10.1`
Finance	`192.168.10.64/26`	`192.168.10.65 – 192.168.10.126`	`192.168.10.127`	`192.168.10.65`

Conclusion: Both departments use separate `/26` subnets. Each subnet supports **62 usable host addresses**, and the Layer-3 router allows communication between the Administration and Finance LANs using their respective default gateways.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
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